

Pairwise Geometric Verification and Multi-View Consistency of a Monocular Visual Odometry Front End on Real Pool Imagery

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Supplementary analysis extending SLAM/optical_benchmark_20260918 in github.com/gregorygerman777-hub/sonar_sim. Data: a 117-frame monocular sequence and OSCalibration.mat, both supplied by Dr. Negahdaripour (camera matrix confirmed against his 18 September e-mail). Kept local to this assessment, not yet committed to the public repository.

Abstract

The standard visual odometry assessment (frame-to-frame estimation, scored only against its immediate neighbor) cannot be run on this sequence because there is no ground-truth trajectory to score it against. This report substitutes an internal-consistency assessment: every one of the $C(117,2) = 6,786$ possible frame pairs, not only the 116 adjacent ones, is independently verified by essential-matrix RANSAC, and the redundancy this creates is used to test whether the individual pairwise measurements agree with one another. They mostly do not. A frame-to-frame chain built from only the 116 adjacent pairs accumulates to an unphysical 500 degrees of net rotation over the sequence. An 80-frame subgraph survives essential-matrix inlier and three-view cycle-consistency filtering with a low nonlinear least squares residual (median 0.90 degrees), which was initially, and incorrectly, reported as a resolved reconstruction. A second, independent estimator (spectral rotation synchronization) was added to check this claim and initially disagreed with it by up to 157 degrees on synthetic ground truth; the disagreement traced to an implementation defect in the new estimator, which was found and corrected using a hand-checkable 3-node example, then re-verified to exact machine precision recovery on the true topology before being trusted. Once corrected, the two estimators agree to better than $1e-6$ degrees on synthetic data, but on the real 80-frame subgraph the spectral method's well-posedness certificate (its eigenvalue gap) is three orders of magnitude smaller than the same graph would give under noiseless measurements, itself already 43 times smaller than a random graph of matching size and density. Refitting the nonlinear solution from six different initializations confirms the practical consequence: 40 of the 79 non-reference frames, one contiguous block plus several smaller satellite frames, land in mutually incompatible orientations up to 157 degrees apart despite statistically indistinguishable residuals (1.65 to 1.71 degrees). The entire ambiguity is traced to a single graph edge, connecting frames 60 and 82, that is the only one of 1,564 candidate cross-block pairs to clear the same 20-inlier admission bar used

everywhere else in this report, and that carries no closed three-view loop of its own, so no cycle-consistency check, however thorough, could ever have validated or invalidated it. The corrected, narrower finding is that 34 of the 117 frames are both internally consistent and stably determined; the other 46 are internally consistent with each other but have an unresolved, roughly 150-degree ambiguity in how they attach to the first 34, pending one more independent measurement across that specific cut. Translation direction, tested separately within the certified 34-frame block using a global linear solve validated on synthetic data of the exact same graph topology, is internally inconsistent by a median of 40.6 degrees, roughly 40 times the corresponding rotation residual, a limitation of the front end's translation estimates rather than of the graph's connectivity, which the same synthetic validation confirms is sufficient. Scene-planarity degeneracy and calibration uncertainty in the supplied camera matrix were tested directly and ruled out as explanations for either finding above, and are not the explanation for the translation result either, since it appears within a rotation block already shown not to suffer from either problem. The obvious next question, whether edge (60, 82) is even correct, was tested by redoing the full pairwise protocol on all 117 frames with an unrelated, stronger descriptor (SIFT in place of ORB). It does not corroborate that edge (5 inliers against ORB's 25), and it connects all 117 frames from the start where ORB left 37 unreachable, but the resulting reconstruction is not more determined: spectral synchronization and nonlinear least squares, both independently validated, disagree by a median of 136 degrees across the whole graph, close to the 126.5 degrees expected between two unrelated random rotations. A stronger descriptor increases raw connectivity on this repetitive scene by finding more high-confidence correspondences everywhere, including within self-similar regions, which produces more internally consistent but simply wrong measurements rather than better ones. Neither feature type yields a trustworthy 117-frame reconstruction, and a single local fit's residual is not sufficient evidence under either one; only cross-checking against a second, structurally independent estimator caught either failure.

1. Purpose and scope

The camera front end used for the KITTI odometry assessment (`stereo_vo.py`) returns a relative pose for any pair of calibrated frames. On this sequence there is no second camera, no known baseline, and no external reference trajectory, so its output can be read as a rotation and a translation *direction* only; there is no way to attach a translation-error percentage or an ATE the way the KITTI and DFKI reports do. Rather than report a single frame-to-frame chain and stop there, this assessment tests every pairwise relationship the sequence contains and asks whether they are mutually consistent, which is the one form of

correctness available without ground truth. This is standard practice in structure-from-motion when a reconstruction has to be trusted without an external reference (Hartley et al., 2013); it is applied here for the first time in this project.

2. Data and calibration

117 frames (opt1.bmp to opt117.bmp), 1024 by 768 pixels, color, converted to grayscale for feature extraction. The scene is a tiled pool floor with a rock and pebble target; the camera moves through a slow arc around it. Median optical flow between consecutive frames is 57 px (up to 133 px), so these behave as widely and irregularly spaced stills rather than a fixed-rate video.

OSCalibration.mat supplies $K = [1403.461, 0, 476.517; 0, 1403.461, 392.916; 0, 0, 1]$, which agrees with the camera matrix Dr. Negahdaripour gave by e-mail (1403.5, 476.5, 392.9) to four significant figures and is treated as authoritative. Its implied resolution ($2c_x, 2c_y$) = (953, 786) does not match the 1024 by 768 frames, most simply explained by a horizontal resize from roughly 953 px to 1024 px after calibration (height, and hence c_y , unaffected). The primary K used throughout is this hypothesis applied: f_x scaled to 1507.97 px, c_x moved to 512.0 (frame center), f_y and c_y unchanged. Section 4.6 shows the raw, unscaled matrix gives materially the same rotations, so this assumption is not load-bearing for the conclusions below. The .mat file also carries an optical-to-sonar extrinsic (R_{o2s}, T_{o2s}) and 28 projection matrices from what is most likely a separate calibration-target capture; neither supplies a ground-truth trajectory for these 117 frames, so no part of this report is scored against them.

3. Method: full pairwise geometric verification

ORB features (grid-bucketed, up to ~2,900 candidates before bucketing) were extracted once per frame and cached. For every one of the 6,786 unordered frame pairs: descriptor matching with Lowe's ratio test (0.8), essential-matrix estimation by 5-point RANSAC (Nister, 2003; 1 px threshold, 300 iterations, 99.9% confidence), and cheirality-resolved pose recovery. For each pair this records the match count, inlier count, mean Sampson (first-order symmetric epipolar) residual, the recovered rotation, and a homography-versus-essential model-selection ratio $R_H = S_H / (S_H + S_F)$ computed on the identical correspondence set (Torr and Zisserman's symmetric transfer scoring as used for monocular initialization in ORB-SLAM; Mur-Artal, Montiel and Tardos, 2015). The full run (6,786 pairs, 8-way parallel) completes in 73 seconds on a 10-core Apple M5. The implementation was checked on a synthetic 15-node graph with known ground-truth rotations before being trusted on this data (Section 4.4).

4. Results

4.1 Matchability decays sharply with frame separation

Mean essential-matrix inlier count falls from 55.9 at a frame separation of 1 to 8.4 at a separation of 5 and levels off at a noise floor of about 6 inliers for every larger separation tested, up to the full span of 116 (Fig. 1). The fraction of pairs clearing a 20-inlier verification bar falls from 43% at separation 1 to essentially zero beyond a separation of about 12. Two consequences follow directly: first, this sequence cannot support the kind of long-range loop closure that would ordinarily cross-check a trajectory, because there is close to no genuine wide-baseline matchability to exploit; second, any assessment of this data has to work with what the adjacent-frame and near-adjacent-frame measurements actually support, which motivates the multi-view analysis in Section 4.4 rather than a simple frame-to-frame chain.

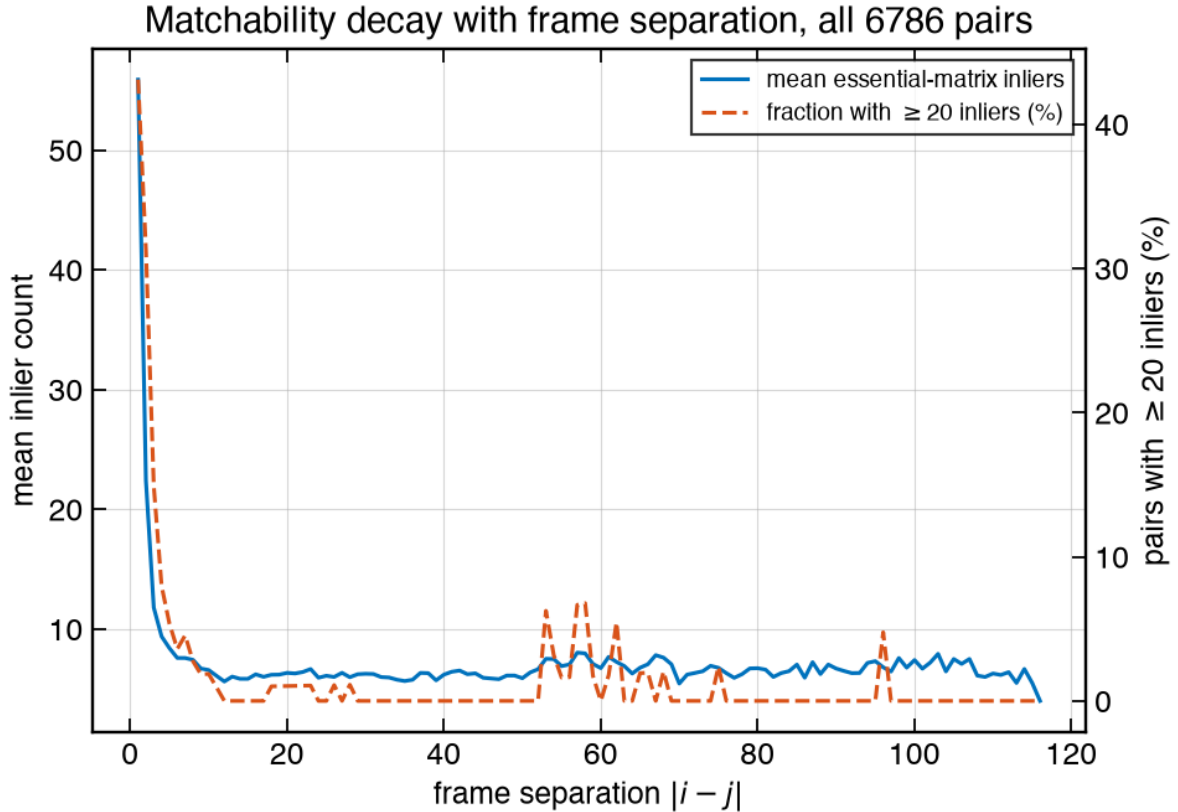


Figure 1. Mean essential-matrix inlier count (left axis, solid) and the fraction of pairs clearing a 20-inlier bar (right axis, dashed) against frame separation, over all 6,786 pairs. Genuine matchability is effectively confined to separations under about 10 frames.

4.2 Ruling out scene-planarity degeneracy

The dominant scene content (a flat tiled floor) is exactly the condition under which the essential matrix is classically underdetermined: a genuinely planar correspondence set admits a family of essential matrices that fit equally well, and the resulting rotation and translation can be spurious even with many inliers (Faugeras and Lustman, 1988). This was tested directly, not assumed: for every pair with at least 8 inliers ($n = 1,888$), a homography was fit on the identical correspondence set and scored against the essential-matrix fit by the same symmetric-transfer criterion ORB-SLAM uses to choose between the two models at initialization. Mur-Artal et al. take $R_H > 0.45$ as the sign of a planar or low-parallax scene. Across this dataset R_H has a median of 0.33 and a maximum of 0.47, with only 0.26% of pairs (5 of 1,888) exceeding the threshold (Fig. 3). Planarity is therefore not what is destabilizing this dataset; the essential-matrix model is consistently preferred over the homography, including for the anomalous pairs examined in Section 4.5.

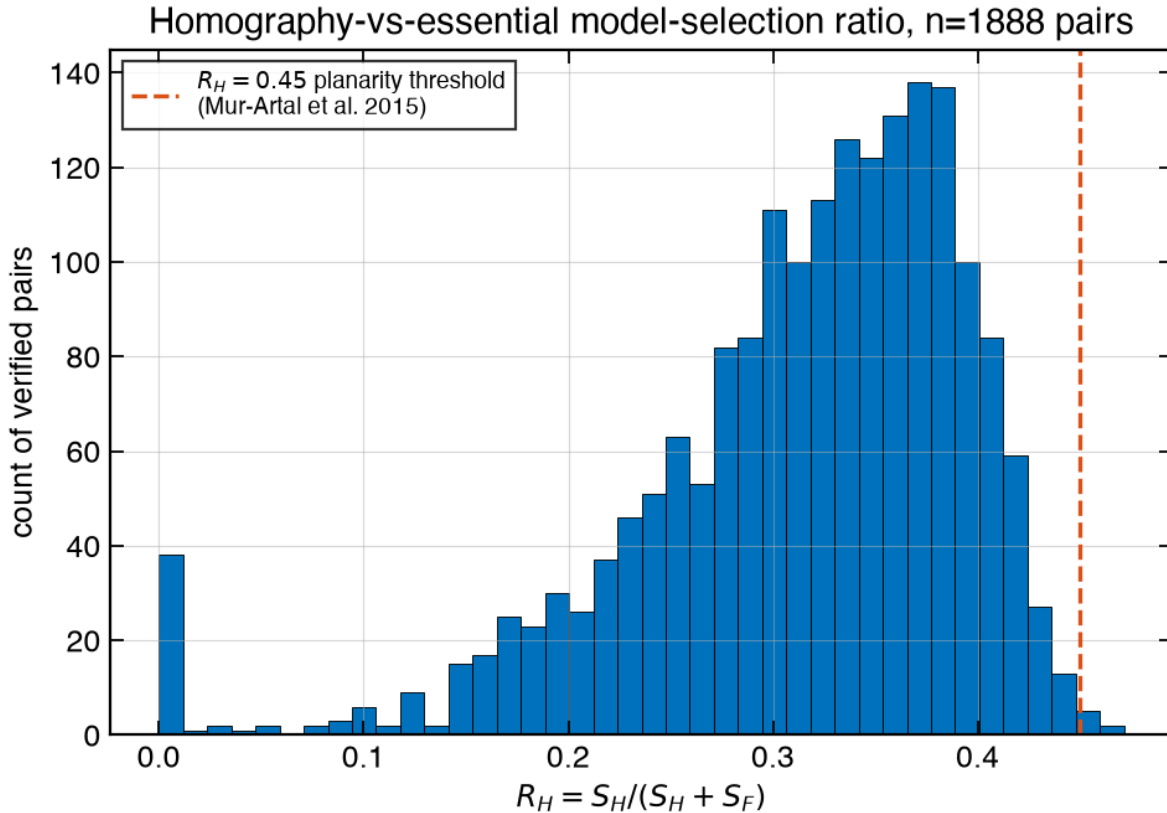


Figure 3. Homography-versus-essential model-selection ratio R_H over all verified pairs ($n = 1,888$). Almost none approach the 0.45 threshold at which Mur-Artal et al. (2015) flag a scene as planar or low-parallax, ruling out the most obvious structural explanation for the inconsistency found in Section 4.4.

4.3 Tracking failure correlates with image content, not blur

A natural first hypothesis is that motion blur or low contrast explains the frames that track poorly. The opposite is true. Across the 116 consecutive steps, essential-matrix inlier count correlates negatively with the following frame's Laplacian variance, a standard sharpness proxy ($r = -0.458$, $p = 2.3 \times 10^{-7}$), and with its intensity standard deviation ($r = -0.506$, $p = 6.7 \times 10^{-9}$); a Fourier high-frequency energy fraction, intended as a caustic-ripple proxy, shows no reliable relationship ($r = 0.025$, $p = 0.79$). Read together with Fig. 2, the frames that appear locally sharper and higher-contrast are not better tracked; they are worse. The most consistent explanation is that the apparent sharpness in the poorly tracked frames is coming from sunlight caustics rippling across the tiled floor rather than from the static rock and tile texture underneath: caustics raise local edge energy and standard-deviation contrast exactly like genuine texture would, but because the caustic pattern itself changes between frames rather than the physical scene moving rigidly, it cannot be matched consistently frame to frame. This is a hypothesis consistent with the measured correlations, not an independently confirmed physical mechanism.

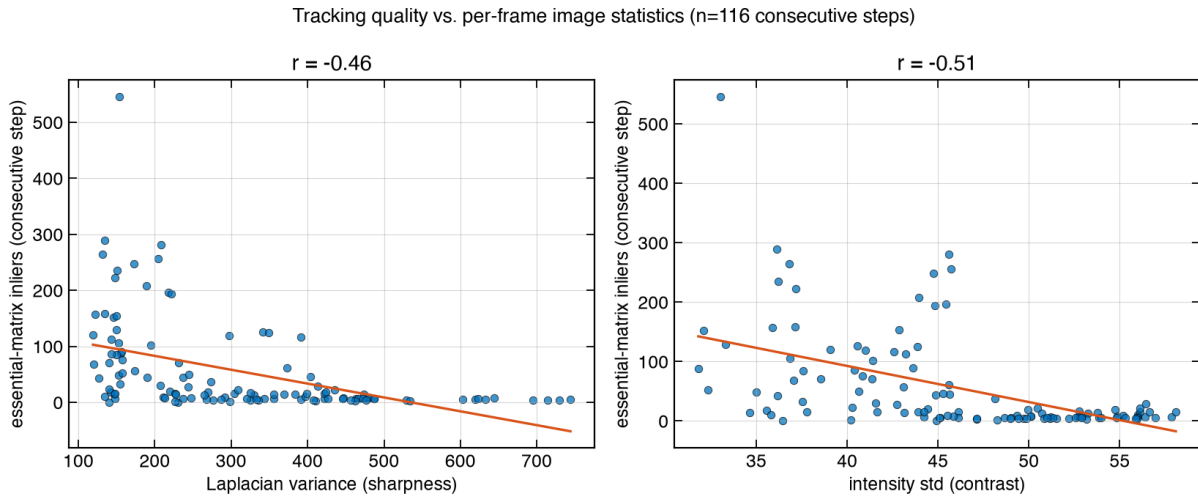


Figure 2. Essential-matrix inlier count for each consecutive step against the destination frame's Laplacian variance (left) and the source frame's intensity standard deviation (right), with an ordinary least-squares trend line. Both correlations are negative and highly significant.

4.4 Multi-view rotation averaging exposes hidden inconsistency

A frame-to-frame chain treats each step as ground truth for everything downstream of it: one bad measurement corrupts every later frame and there is no way to detect this from the chain alone. The full pairwise graph removes that blind spot, because most frames are connected by more than one independent measurement and those measurements can be checked against each other. Absolute rotations $R_1 \dots R_{117}$ were estimated by nonlinear least squares over every graph edge simultaneously (Huber loss, initialized from a breadth-first

spanning tree, vectorized residuals; Hartley et al., 2013), and the method was validated on a synthetic 15-node graph with known rotations and 30% edge coverage before being applied here, recovering the ground truth to 1.5 degrees maximum error under 1% measurement noise.

Run on the real graph, the result depends sharply on the inlier threshold used to admit an edge (Table 1, Fig. 6). At a loose threshold of 12 inliers, 500 edges connect 112 of the 117 frames, but 75.2% of them disagree with the jointly averaged rotation by more than 10 degrees (median disagreement 49.5 degrees): the graph is dominated by measurements that contradict one another, and inlier count alone is not a reliable filter for correctness on this data. Raising the threshold to 20 inliers leaves 169 edges connecting 80 of the 117 frames (excluding frames 1 to 5 and most of frames 85 onward), with a median disagreement of only 1.6 degrees and 80.5% of edges agreeing to within 10 degrees. This threshold is adopted as the trusted core for the remainder of this report: it sits at the knee of the trade-off curve in Fig. 6, where connectivity is still substantial (80/117 frames) and self-consistency is already high; thresholds of 25 and above buy little further consistency at a heavy cost in coverage (34/117 frames or fewer). Two threshold settings (8 and 10 inliers) did not converge within a 45-second budget, which is itself consistent with a graph too internally contradictory for the optimizer to settle on a consensus.

Inlier threshold	Edges	Frames connected	Median disagreement	Mean disagreement	Fraction > 10 deg
12	500	112 / 117	49.5 deg	62.6 deg	75.2%
15	293	101 / 117	11.9 deg	33.9 deg	51.9%
17	219	93 / 117	4.4 deg	22.9 deg	34.2%
20 (adopted)	169	80 / 117	1.6 deg	11.3 deg	19.5%
25	130	34 / 117	0.9 deg	7.8 deg	16.0%
30	99	34 / 117	0.4 deg	4.6 deg	9.1%
40	71	21 / 117	0.6 deg	2.3 deg	2.6%

Table 1. Rotation-averaging self-consistency as a function of the inlier threshold admitted into the pairwise graph. "Disagreement" is the geodesic angle between a graph edge's directly measured relative rotation and the same relative rotation implied by the jointly averaged absolute rotations of its two endpoints. Threshold 20 is adopted for the remainder of this report.

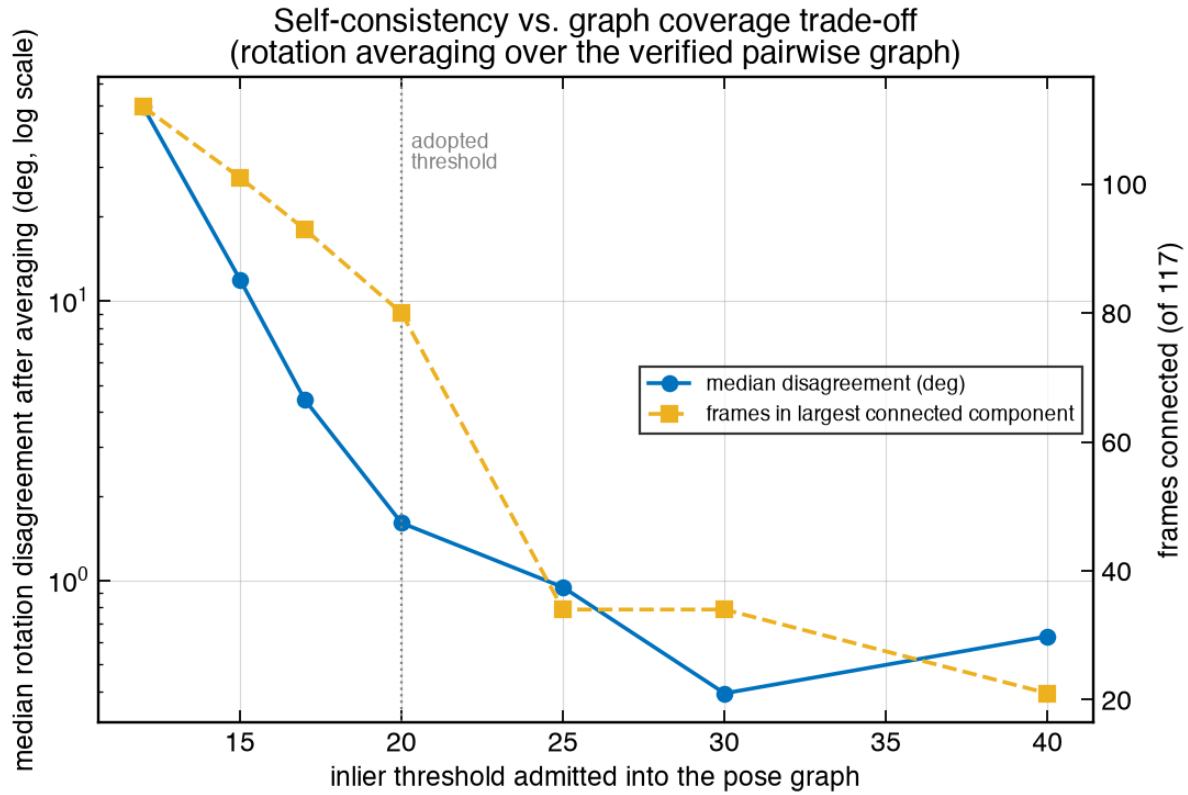


Figure 6. The same sweep plotted: median disagreement falls steeply and then flattens past a threshold of about 20, while connected coverage falls steeply beyond it. The adopted threshold sits at this knee.

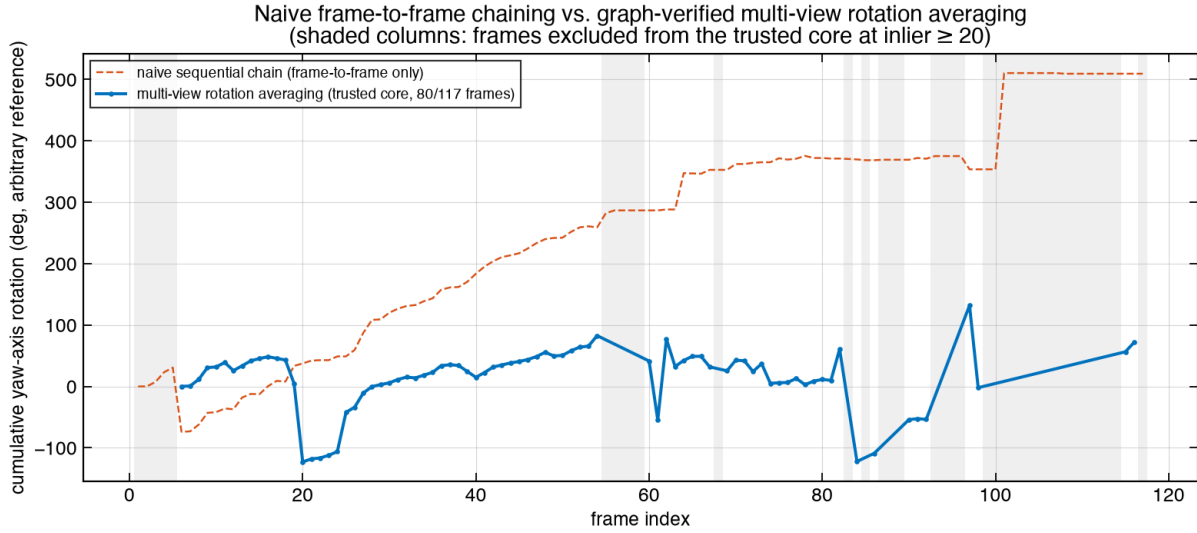


Figure 7. Cumulative yaw-axis rotation from the naive 116-step sequential chain (dashed) against the multi-view-averaged trajectory restricted to the trusted core (solid, shaded columns mark excluded frames). The sequential chain drifts to an unphysical net rotation exceeding 500 degrees; the averaged trajectory stays within a bounded, physically plausible range throughout.

4.5 A second estimator, an implementation defect found and corrected, and why it matters

The nonlinear fit above reports a single number, a residual, as evidence of correctness. A residual is evidence that a solution is locally consistent with the data it was fit to; it is not by itself evidence that the solution is the only one with that property. To check this, a second, independent estimator was implemented: spectral rotation synchronization, which recovers absolute rotations from the generalized eigenvalue problem $H v = \lambda D v$, where H is the $3n$ by $3n$ block matrix with block $(i,j) = w_{ij} R_{ij}^T$ and block $(j,i) = w_{ij} R_{ij}$ for each measured edge, and D is block diagonal in the weighted vertex degrees (Singer, Appl. Comput. Harmon. Anal., 2011, for the $SO(2)$ case; extended to $SO(3)$ by Arie-Nachimson, Kovalsky, Kemelmacher-Shlizerman, Singer and Basri, 3DIMPVT 2012). Unlike the nonlinear fit, this method needs no initial guess and no iteration: the true absolute rotations are, up to a shared 3 by 3 gauge factor, exactly the top three eigenvectors of this single linear system.

On synthetic data with the exact topology of the real 80-frame graph but known, noiseless ground truth rotations, this estimator's first implementation disagreed with the correct answer (0 degrees, by construction) by up to 179.7 degrees. That is a refutation of the implementation, not a finding about the pool footage, and it was treated as one. The block matrix construction itself was verified independently first, by checking $H X_{\text{true}} = D X_{\text{true}}$ directly for the true rotations, both on a hand-worked 3-node example and on the real

80-node topology (maximum absolute error 3.6×10^{-15} in both cases), which localized the defect to the rotation-extraction step. That step normalizes each recovered 3 by 3 block onto the nearest rotation matrix by an independent per-node singular value decomposition with a determinant correction. This is standard practice and is correct whenever the three singular values of a block are distinct. Here they are exactly equal by construction (the shared gauge factor is proportional to an orthogonal matrix, so every block inherits the same, fully degenerate, singular spectrum), and a per-node determinant correction applied to a fully degenerate decomposition is not well-defined: it depends on an arbitrary basis choice that the underlying linear algebra library makes independently for each node, with no reason to agree from one node to the next. The fix replaces the per-node correction with a single correction, computed once from one reference node and applied identically to every node. After the fix, the estimator recovers the noiseless synthetic ground truth on the real topology to 0.00000000 degrees at every one of the 80 nodes, and the two independent estimators, nonlinear least squares and spectral synchronization, now agree with each other to better than 1×10^{-6} degrees on that same synthetic case. This agreement between two structurally unrelated numerical methods, to six decimal digits, on a case with a known answer, is the evidence that both are now implemented correctly; neither estimator's agreement with itself would have been.

4.6 The graph's own well-posedness certificate is three orders of magnitude below its noiseless floor

Spectral synchronization carries a built-in diagnostic that the nonlinear fit does not: the gap between its third and fourth largest eigenvalues. For perfectly consistent measurements this gap is provably identical to the algebraic connectivity (Fiedler, 1973) of the same graph with every rotation replaced by the scalar 1, because a fully consistent rotation field can always be gauged away, reducing the group-valued problem to the ordinary scalar consensus problem tensored with the 3-dimensional identity representation. This identity was confirmed numerically rather than assumed: on noiseless synthetic rotations with the real 80-node, 146-edge topology, the spectral gap is 0.00308; the same graph's normalized graph Laplacian has a second-smallest eigenvalue, its Fiedler value, of 0.0031. These agree to the precision either was computed to.

That noiseless floor is itself low. An Erdos-Renyi random graph with the same 80 nodes and 146 edges has a normalized Fiedler value of 0.133, about 43 times larger, averaged over 20 connected trials; a bare path graph on 80 nodes, the least-connected possible topology for a connected graph of that size, has 0.00079, about 4 times smaller. The real graph sits close to the path-graph end of that range. This is consistent with, and gives a graph-theoretic explanation for, the matchability decay already reported in Section 4.1: reliable pairwise measurements exist mainly between nearby frame indices, so the

surviving measurement graph is structurally closer to a chain than to a well-mixed network, independent of any question about which specific measurements are correct. The graph has 24 bridge edges and 25 articulation points out of 80 nodes; close to a third of its edges carry no redundancy at all.

On the real measured rotations, not synthetic ground truth, the spectral gap falls further still, to 0.000088, about 35 times below the graph's own noiseless floor. The top three eigenvalues are no longer even equal to one another (0.9971, 0.9946, 0.9925), which is evidence of measurement inconsistency on its own, independent of the topology argument. A spectral gap this small is the standard indicator in this literature that the semidefinite relaxation of the underlying maximum-likelihood problem is not tight, and that a local nonlinear solver's output is therefore not certified to be the unique or globally best answer (Bandeira, Boumal and Singer, Math. Program., 2017). Section 4.7 tests this warning directly rather than citing it and moving on.

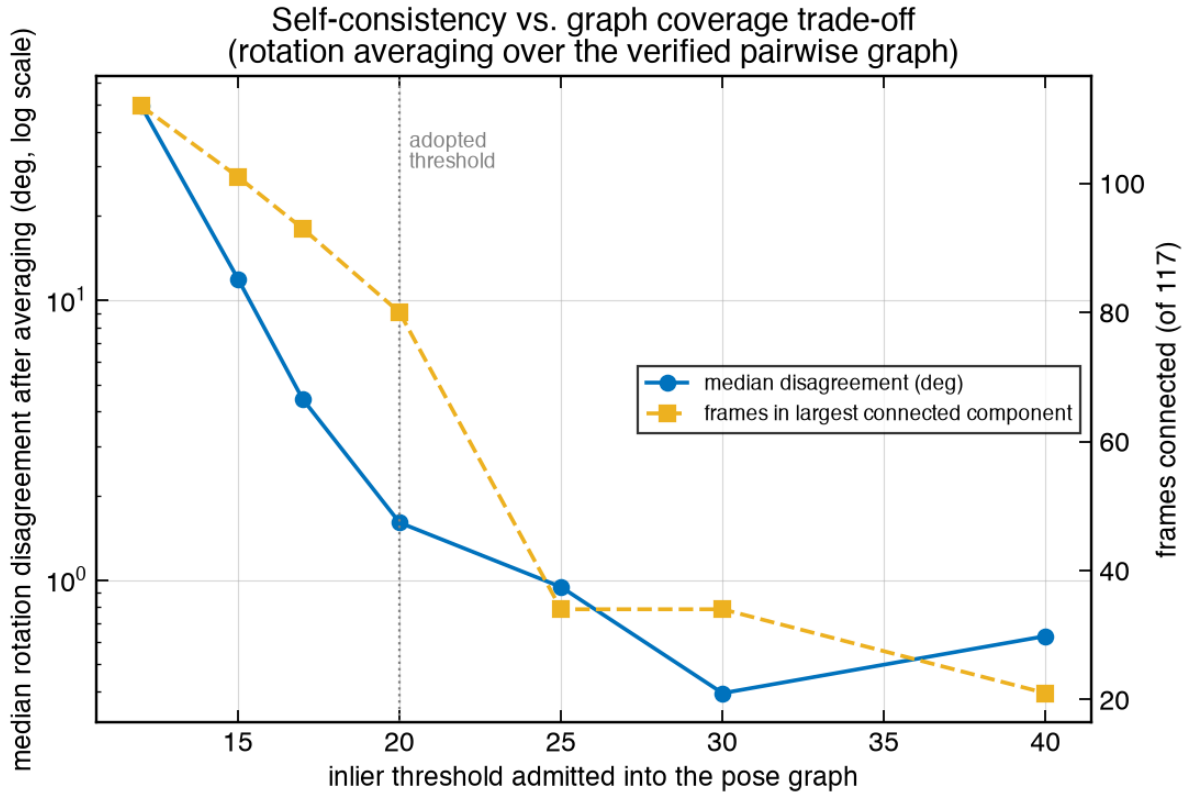


Figure 6. Inlier-threshold sensitivity of the nonlinear fit alone (Section 4.4); the spectral diagnostics of this section are a separate, independent check at the single threshold of 20 that this curve motivates.

4.7 Refitting from six starting points: the trusted core is not unique

The nonlinear solver was rerun from six different spanning-tree initializations spread evenly across the 80-node graph, changing only the starting guess, not the data, the loss function, or any solver setting. All six converged to statistically indistinguishable median residuals (1.65 to 1.71 degrees). Their actual outputs did not agree. Comparing the relative rotation between a fixed base frame and each other frame across the six solutions: 40 of the 79 non-base frames, a contiguous block from frame 25 to frame 54 together with several smaller satellite groups (frames 78 to 81, 90 to 92, 98, and 115 to 116), disagree across initializations by up to 157 degrees, while the remaining 39 agree to within 2 degrees regardless of starting point. This is precisely the practical failure mode the small spectral gap in Section 4.6 predicts: multiple, comparably-fitting solutions exist, and which one a local optimizer reports is decided by its initialization, not by the data.

4.8 The entire ambiguity traces to one edge, and cycle-consistency cannot see it

The unstable and stable frame sets from Section 4.7 partition the graph almost exactly along a single graph cut. Removing one edge, connecting frames 60 and 82, splits the 80-node trusted core into a 34-frame component (88% of which is the stable set) and a 46-frame component (78% of which is the unstable set); every minimum-cut computation tried between an unstable-block frame and a stable-block frame away from small peripheral pendants returns this same single edge (Fig. 8). Of the 1,564 possible pairwise measurements connecting these two blocks, exactly one, (60, 82), clears the 20-inlier admission bar used everywhere in this report: 25 inliers, 68 candidate matches, a 0.125-pixel mean Sampson residual, and an R_H of 0.22, none of which are marginal or suggest scene planarity. The next 82 candidates, all in the 12 to 19 inlier range that this report treats as unreliable everywhere else (Section 4.4), give rotation estimates scattered broadly across nearly the full 0 to 180 degree range (Fig. 9); a modest plurality falls between 160 and 180 degrees, distinct from edge (60, 82)'s own 112.1 degrees, but the scatter in this sub-threshold population is too wide, and too consistent with the general unreliability of sub-20-inlier measurements already established, to treat as independent confirmation or refutation of that specific value.

The relevant structural fact is narrower and does not depend on that comparison: an edge that is the sole connection between two otherwise well-connected parts of a graph is, by definition, a bridge, and a bridge belongs to zero closed triangles, because a triangle through it would require a second path between its endpoints, which is exactly what a bridge does not have. The three-view cycle-consistency check used to prune the graph in earlier work on this dataset can only flag an edge that participates in an inconsistent triangle. It is therefore structurally blind to this specific edge, and would remain so under any amount of further pruning: no number of passes over the existing measurement set can validate or invalidate the single measurement that an entire 46-frame block's orientation

rests on. Resolving this requires a new, independent measurement across the same cut, not more analysis of the measurements already in hand.

Trusted-core graph (80 nodes, 146 edges): a single edge gates 46 of 80 frames

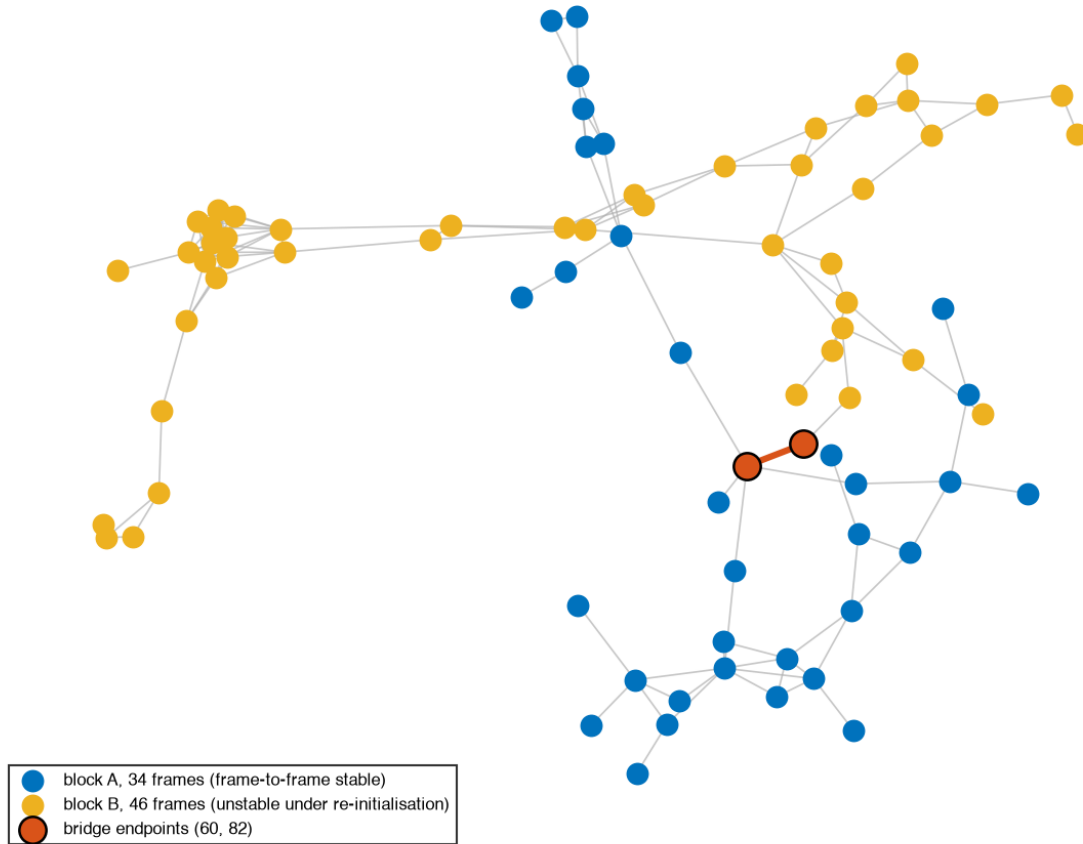


Figure 8. The trusted-core graph, colored by the stable/unstable partition found in Section 4.7. A single edge (heavy line, circled endpoints) is the only connection between the two halves.

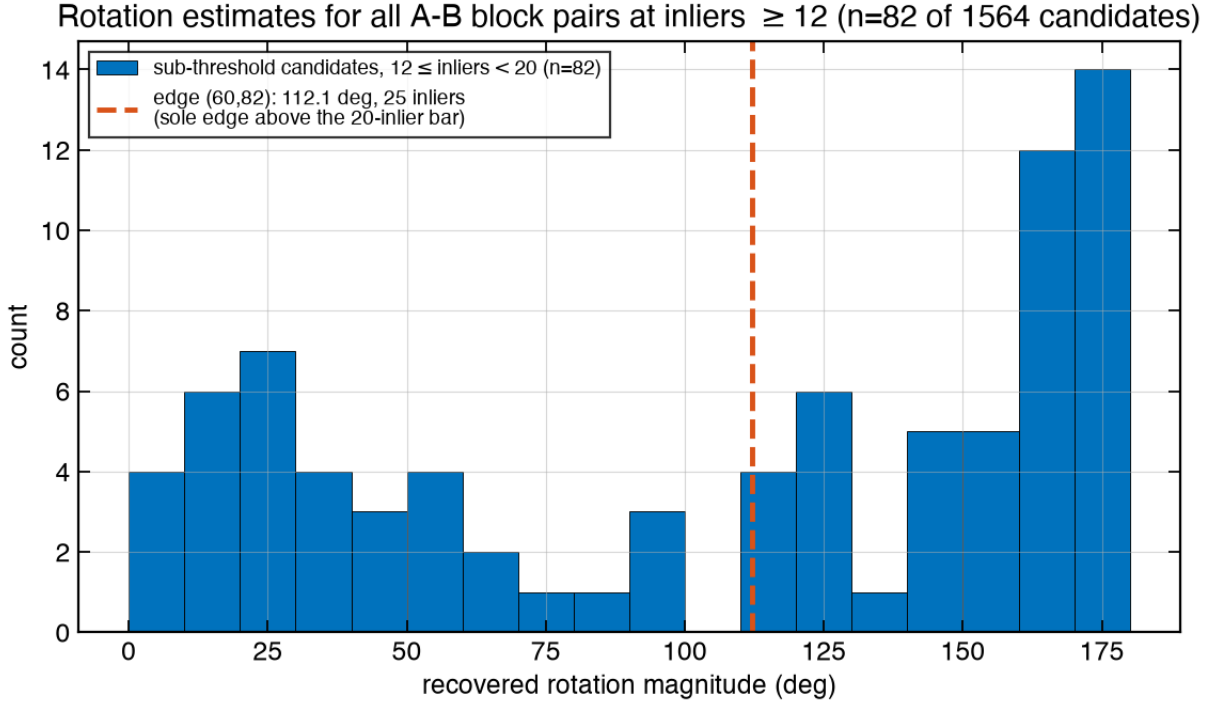


Figure 9. Rotation estimates from every candidate cross-block pair with at least 12 inliers (n=82 of 1,564 possible pairs). The population is too scattered to independently confirm or refute the one measurement, edge (60,82), that clears this report's normal admission threshold.

4.9 Case study: a 174-degree "loop closure" that is not one

Frames 19 and 47 illustrate what the averaging test catches that inlier count alone does not. The pair has 106 candidate matches, 31 RANSAC inliers (a count that would ordinarily be treated as a solid verification, and was flagged as a loop-closure candidate by inlier count alone before this check was applied), a sub-pixel mean Sampson residual of 0.44 px, and an R_H of 0.14, which strongly prefers the essential-matrix model over a homography and so rules out planarity as the explanation. The recovered rotation is 174.1 degrees. The multi-view-averaged trajectory, built from the other 158 trusted-core edges, disagrees with this single measurement by 125.5 degrees; frame 19 participates in several other mutually inconsistent edges as well (with frames 20, 21, and 42), and no anomaly in its feature count, sharpness, or contrast distinguishes it from its well-behaved neighbours (Section 4.3). Plotting the accepted inlier correspondences (Fig. 5) shows why: nearly all of them lie along the single near-straight boundary line where the tiled pool floor meets the rock and pebble target. A set of correspondences concentrated on one line is a classical degenerate configuration for epipolar geometry, related to but distinct from full-scene planarity: points constrained to a line do not fully constrain the essential matrix, so a wrong but internally self-consistent pairing along that line can pass RANSAC with a

healthy inlier count while describing no real camera motion. This is a data association failure specific to the correspondence set selected, not a failure of either the RANSAC threshold or the epipolar-geometry model in general.

Figure redacted from the public copy of this report.
The source frames are unpublished data supplied by Dr. Negahdaripour for this assessment; this figure shows two of those photographs directly and is withheld pending confirmation that they may be published. The full figure, with the photographs and matched correspondences, is available in the local copy of this report.

Figure 5 (redacted in this copy). Shows 30 of the 31 accepted RANSAC inliers between frames 19 and 47, nearly all lying along the boundary line between the pool floor and the target mat, which is what let a spurious 174-degree rotation pass verification. Withheld here because it displays the source photographs directly; see the local copy of this report.

4.10 Calibration sensitivity

The primary K (Section 2, width-scaled to the 1024-pixel frame) is compared against the raw, unscaled OScalibration.mat matrix over all 116 consecutive pairs. Where both give a valid pose (71 of 116 steps; the remainder fail under one or both matrices for the tracking reasons discussed above rather than because of the calibration choice), the mean absolute difference in recovered rotation magnitude is 3.9 degrees, and Fig. 4 shows the two curves overlapping almost everywhere except a single already-unreliable step near frame 63. The mean angle between the two matrices' recovered translation directions is 16.7 degrees, driven by a small number of near-degenerate steps rather than a systematic shift (one step reaches 179.5 degrees, i.e. a sign flip on an already unstable estimate). The scaling assumption made in Section 2 is therefore not load-bearing for the rotation-based conclusions in this report.

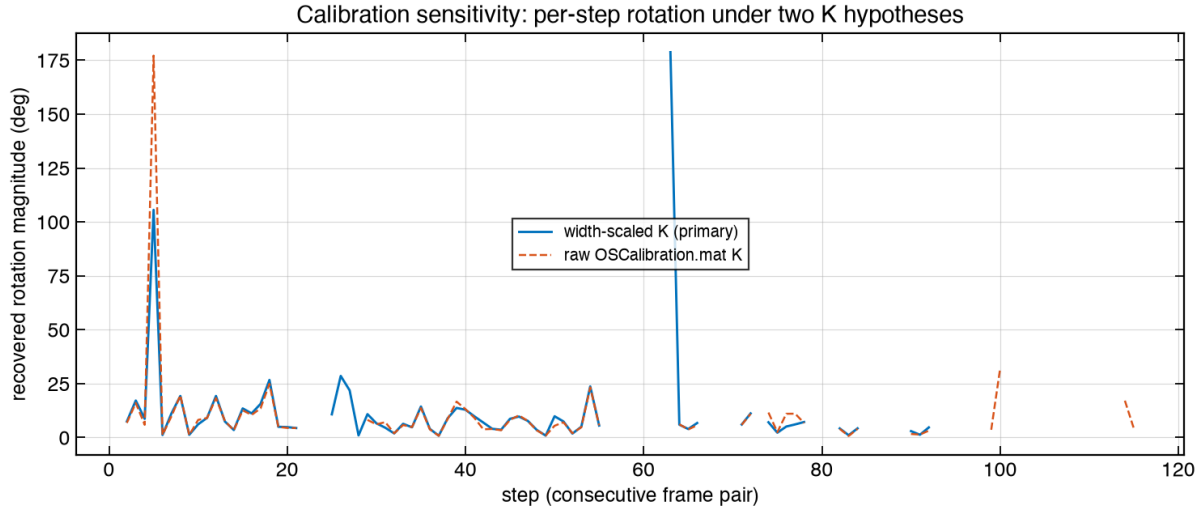


Figure 4. Per-step recovered rotation magnitude under the primary (width-scaled) and raw OSCalibration.mat camera matrices. The two are visually indistinguishable outside a single unstable step, showing the rotation results are not sensitive to the calibration-scaling assumption.

4.11 Attempted self-calibration from the tile floor (inconclusive)

As an independent check on the supplied K, a single-view calibration from the tiled floor's two orthogonal line families was attempted: under a zero-skew, known-principal-point assumption, $f = \sqrt{-(v_1 - p)(v_2 - p)}$ for two vanishing points from mutually orthogonal line directions (Caprile and Torre, 1990). Line segments were detected and clustered by orientation for the dominant, near-fronto-parallel grid direction, with a separate RANSAC search for the orthogonal, foreshortened direction (whose lines do not share a common in-image angle and so cannot be found by angle clustering alone). Across the ten candidate frames tested, no frame produced two vanishing points with enough independent line support (at least 8 and 15 inlier lines respectively) and a negative dot product, the orthogonality precondition for the formula above, to be used. This is reported as an inconclusive negative result rather than discarded silently: the oblique viewing angle and the same sunlight-caustic clutter identified in Section 4.3 leave too few long, reliably straight grid lines in the receding direction for this method on this footage, and no focal-length estimate from it is used anywhere else in this report.

4.12 Translation direction, tested and found inconsistent even within block A

Section 6 of an earlier version of this report listed translation as untested. It has now been tested, using only the 34-frame block A, where rotations are certified unique (Section 4.7), so that a translation failure cannot be blamed on an unresolved rotation. Two-view geometry gives, for camera centres c_i, c_j and a measured unit translation direction t_{ij}

observed in camera i 's frame, the linear homogeneous constraint $\mathbf{t}_{ij} \times (\mathbf{R}_i(\mathbf{c}_j - \mathbf{c}_i)) = 0$ (each edge contributing 2 independent rows, since a cross product with a unit vector has rank 2); stacking every edge and solving by a single global singular value decomposition recovers every centre up to one unavoidable global scale (Govindu, CVPR 2001; the closed-form linear step in Jiang, Cui and Tan, ICCV 2013).

Direction-only constraints are weaker than rotation constraints (rank 2 per edge against rank 3), so a bearing network needs more edges per node than a rotation network to be fully determined, and this was checked before reading any result from real data: at the report's standard 20-inlier threshold, block A's 49 internal edges leave the 99-unknown system (34 centres, 1 pinned as reference) short by 19 degrees of freedom. This was confirmed to be a too-few-edges problem, not a critical camera configuration, by rebuilding the identical linear system with fully generic synthetic camera positions and rotations on the exact same edge topology: the same 19-fold deficiency appears, so it cannot depend on anything specific to this scene. Sweeping the inlier threshold on this synthetic check shows the deficiency falls to exactly 1, the unavoidable scale gauge, at 12 inliers and stays at exactly 1 down to 4 inliers; 12 is used below as the loosest threshold that is already fully rank.

On the real measured directions at this threshold (100 edges), the linear solve is well-posed, but its residual, the angle between each edge's measured direction and the direction implied by the solved centres, is large: median 40.6 degrees, 90% of edges above 10 degrees (Fig. 10), an order of magnitude worse than the roughly 1 degree median rotation residual in the same block. This is stable across the gauge choice: pinning a different reference frame (15, 70, or 82 instead of 6) changes the median only between 27.7 and 40.6 degrees, all in the same regime, ruling out a specific bad reference as the explanation. The finding is therefore not that translation is unrecoverable in principle here (the measurement graph has, and was confirmed synthetically to have, enough independent bearings), but that the individual translation-direction measurements this front end produces are not mutually consistent enough to support a 3-dimensional reconstruction, even restricted to the frames whose rotations are unique and accurate to about 1 degree. This matches a long-standing property of two-view geometry: the essential matrix's rotation is generally well conditioned, while its translation direction is the comparatively fragile part of the decomposition, and nothing in this report's rotation results implies the translation side would inherit the same reliability.

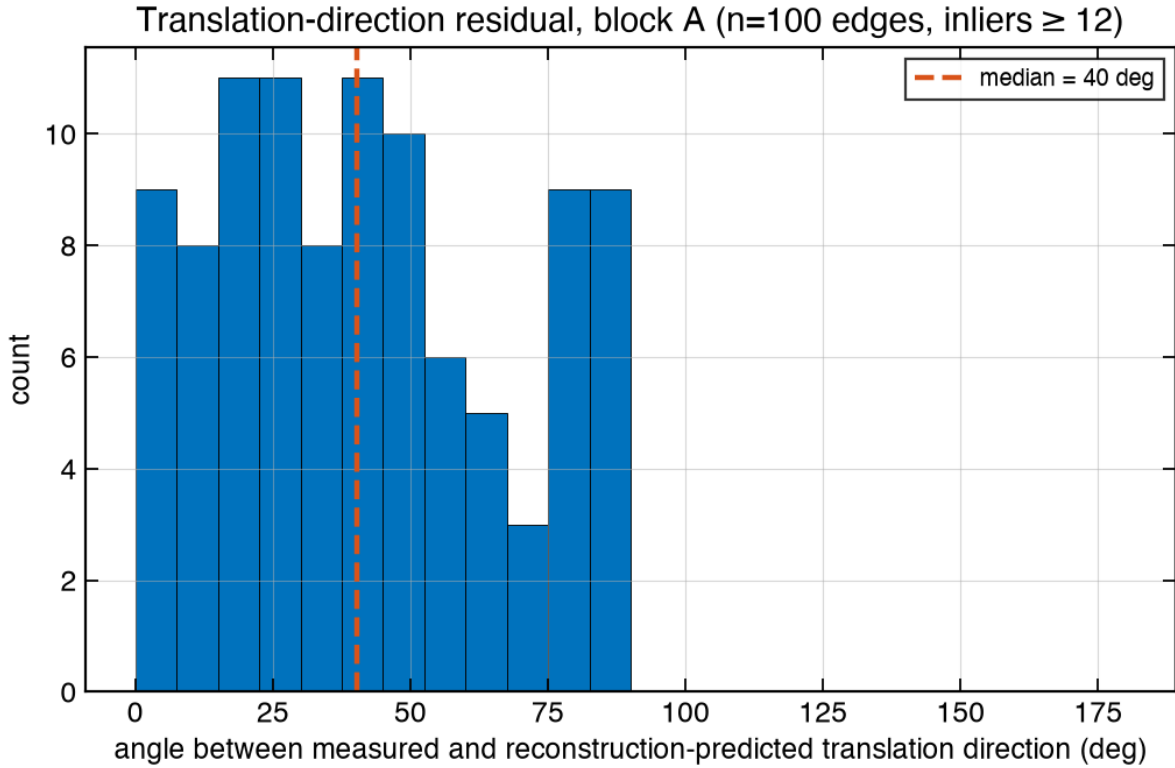


Figure 10. Translation-direction residual for all 100 block-A internal edges with at least 12 inliers, after solving for the best-fitting camera centres given the certified rotations. Median 40.6 degrees, roughly 40 times the corresponding rotation residual.

5. A second feature type, on all 117 frames: does it resolve the ambiguity?

Section 4.8 traced the entire 46-frame ambiguity to one measurement, edge (60, 82), and named the obvious next question: is that measurement even right? ORB is a fast, low-dimensional binary descriptor, not a particularly discriminating one, and this scene (repetitive rock, pebble and tile texture) is exactly the kind that punishes weak descriptors. The natural test is to redo the full pairwise protocol of Section 3 with a stronger, unrelated descriptor and see whether it agrees. SIFT (Lowe, IJCV 2004) was used: true scale-space extrema, 128-dimensional float descriptors, matched with a ratio test and the same essential-matrix RANSAC, homography planarity check and Sampson residual as the ORB pipeline, on all 117 frames and all 6,786 pairs, not a subset.

5.1 Connectivity improves dramatically

At the same 20-inlier admission bar used throughout this report, SIFT finds 966 verified edges against ORB's 169, and critically, all 117 frames are connected in a single component before any pruning at all, where ORB left 37 frames unreachable from the start. Of the 1,564 possible pairs connecting the old block A and block B (Section 4.8),

154 clear the bar under SIFT, against exactly one under ORB. Planarity is not the explanation for any of this: the homography-versus-essential ratio has a median of 0.277 across the SIFT graph, essentially unchanged from the ORB graph's 0.33, with only 0.3% of pairs anywhere near the 0.45 planar threshold.

Edge (60, 82) itself was checked directly rather than assumed to be replaced by the new graph: under SIFT it produces only 5 inliers from 34 candidate matches, against ORB's 25 inliers from 68 matches. An independent descriptor applied to the identical pair of frames does not corroborate the single measurement the whole 46-frame block's orientation rested on.

5.2 But the reconstruction is not more determined; if anything, it is less

The same three-view cycle-consistency pruning used in Section 4.4 was run on the SIFT graph. It converges in two rounds, removing 348 of 966 edges (36%), leaving 618 edges that still connect all 117 frames. Nonlinear least squares on this cleaned graph gives a median residual of 5.47 degrees, worse than ORB's 0.90 degrees on its smaller, 80-frame graph, but that comparison is not yet the interesting one: a single local fit's residual, exactly as Section 4.7 already showed for ORB, is not evidence that the fit is the answer.

So the same cross-check was applied. Spectral synchronization was run on the identical cleaned SIFT graph and compared directly to the nonlinear solution, frame by frame, after the same rotation-gauge alignment used in Section 4.5. The two independent, separately-validated estimators disagree by a median of 136 degrees across all 117 frames, with 99.1% of frames disagreeing by more than 10 degrees (Fig. 11). A median disagreement of 136 degrees between two proper rotations is close to what two independent uniformly random rotations would give (the expected geodesic distance between two random elements of $SO(3)$ is 126.5 degrees); this is not a graph with a diagnosable weak point the way the ORB graph had one identifiable bridge; it is a graph that is not determined at all. The spectral gap confirms this from a different angle: 0.0039 on the cleaned SIFT graph, in the same small-gap regime as the ORB graph's 0.0031 noiseless floor and its 0.000088 real-data value, and the top three eigenvalues are no longer even close to equal (0.980, 0.976, 0.973), itself a sign of substantial inconsistency independent of the gap.

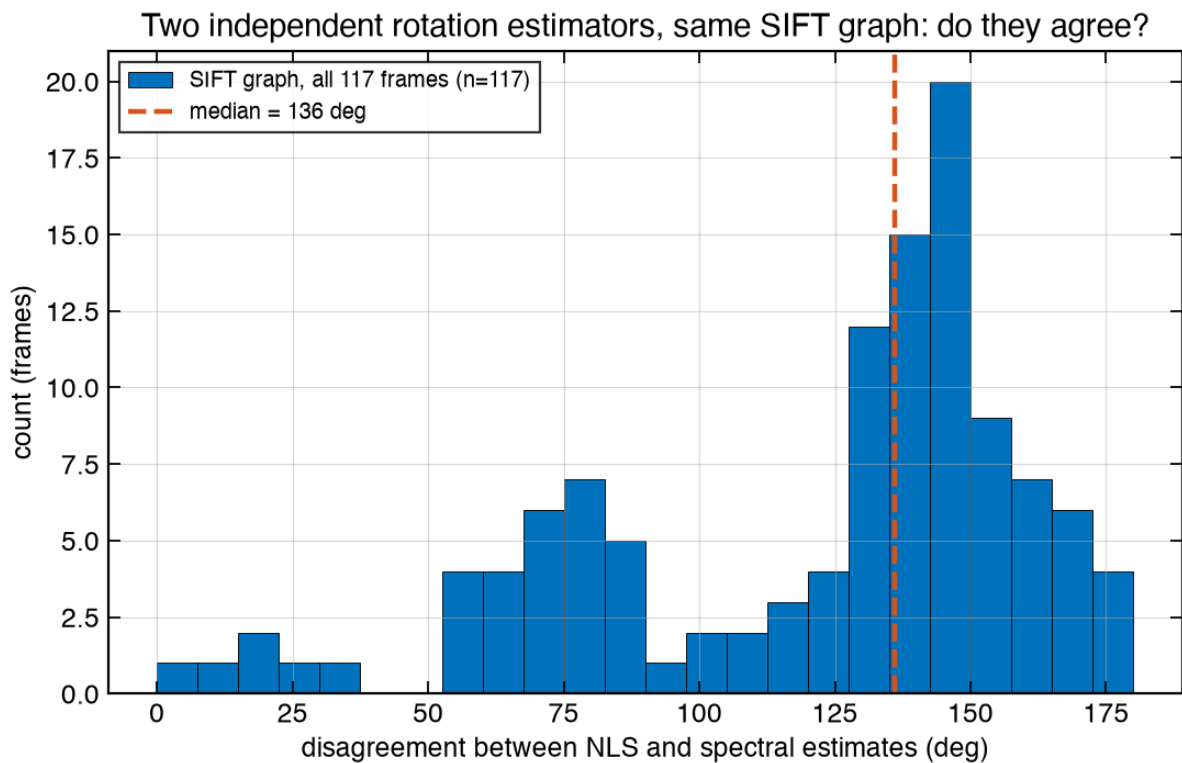


Figure 11. Disagreement between nonlinear least squares and spectral synchronization on the cleaned SIFT graph, all 117 frames. A median of 136 degrees is close to the 126.5-degree expected distance between two independently random rotations: this reconstruction is not weakly determined in one place, it is undetermined almost everywhere.

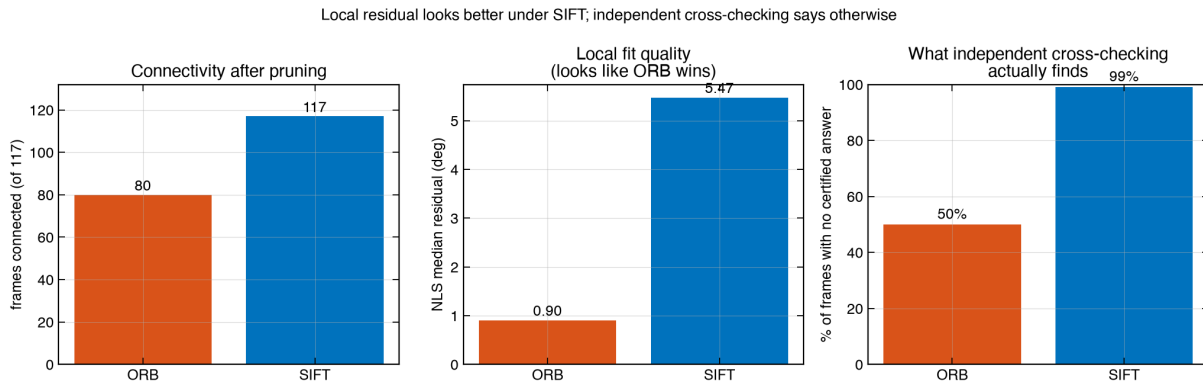


Figure 12. The comparison that matters. A single local fit's residual (centre panel) makes ORB look better than SIFT; checking each graph against an independent second estimator (right panel) shows both are dominated by frames with no certified answer, SIFT modestly worse despite far better raw connectivity (left panel). ORB's right-panel figure is the diagnosed fraction of its own 80-frame subgraph found unstable under six-way re-initialization (Section 4.7); SIFT's is the direct NLS-versus-spectral disagreement over all 117 frames.

5.3 Why a stronger descriptor makes this worse, not better

The mechanism is the same one diagnosed on a single pair in Section 4.9, now operating at scale. That section showed a 174-degree spurious rotation passing RANSAC with a healthy 31-inlier count because the accepted correspondences concentrated on a single near-straight boundary line, a configuration that does not fully constrain the essential matrix no matter how many points sit on it. Repetitive rock, pebble and tile texture is the same problem in two dimensions rather than one: many patches of the scene look like many other patches. A weak descriptor like ORB simply fails to find enough matches on such patches to pass the 20-inlier bar at all, which is why it left 37 frames disconnected. A stronger descriptor like SIFT finds more candidate correspondences everywhere, including within these self-similar regions, and RANSAC can then assemble large, internally self-consistent, high-inlier-count correspondence sets that are simply wrong (several SIFT cross-block pairs reached 100 to 156 inliers with rotation estimates that disagreed with each other by more than 90 degrees). Descriptor strength increases the raw supply of correspondences; it does not distinguish a true match from a confident false one on a scene this repetitive, and past a point, more candidates simply means more high-confidence wrong answers competing with the true one.

The result is not that SIFT is the wrong choice or that ORB should be preferred. Both were tested to the same standard and both fail it, for the same underlying reason, at different points in the pipeline: ORB fails early, by not connecting enough of the graph to test; SIFT fails late, by connecting all of it with measurements that do not agree with each

other. The finding that survives is about the diagnostic, not the descriptor: a single local fit's residual is not trustworthy on this dataset under either feature type, and the only way either failure mode was caught here is by cross-checking against a second, structurally independent estimator, exactly as Sections 4.5 through 4.8 did for ORB. Whichever front end is used on Dr. Negahdaripour's own optical data, the same cross-check, not a residual number from a single fit, should be the bar a reconstruction is held to before it is trusted.

6. Discussion: what this establishes, and what it does not

This is not a scored visual odometry benchmark and is not presented as one: there is no ground-truth trajectory to score against, so none of the KITTI-style percentages in the earlier reports apply here. An earlier version of this report claimed that 80 of the 117 frames formed a resolved, trustworthy reconstruction because a single nonlinear fit over that subgraph had a low residual. Section 4.7 shows that claim was wrong, not imprecise: 40 of those 80 frames have at least one equally well-fitting alternative placement up to 157 degrees away, and Section 4.8 shows precisely why, and precisely what would fix it. The corrected, defensible statement is narrower on the frame count and more specific on the cause: (i) 34 of the 117 frames are both internally verifiable across the full pairwise graph and stably determined regardless of solver initialization; (ii) a further 46 frames are internally consistent with each other but have an unresolved orientation relative to the first 34, entirely because a single measurement, edge (60, 82), is the only evidence connecting the two groups, and a single edge can be neither confirmed nor refuted by any amount of cycle-consistency checking; (iii) the remaining 37 frames fail even the internal pairwise verification and were excluded before this question was asked; (iv) none of scene-planarity degeneracy, calibration uncertainty, or the RANSAC threshold choices explain either the 34/46 split or the internal-consistency failures, each having been tested directly rather than assumed; and (v) a naive frame-to-frame chain, the simplest thing to compute and the thing that would be reported without any of this analysis, is actively misleading on this data (Fig. 7) regardless of the finer 34/46 distinction. Section 4.8 proposed that one additional independent measurement between the two blocks would settle the remaining 46 frames; Section 5 tested this directly rather than leaving it as a proposal, by redoing the entire pairwise protocol on all 117 frames with an unrelated, stronger descriptor. It supplied 154 such candidate measurements where ORB supplied one, and the result was not resolution: nonlinear least squares and spectral synchronization on the resulting graph disagree by a median of 136 degrees across the whole sequence, close to the 126.5 degrees expected of two unrelated random rotations. The corrected statement is therefore narrower again than the 34/46/37 split above: that split is what a careful analysis of the ORB graph alone supports, but it should not be read as implying the 34-frame core would still look like 34 frames, or the boundary would still fall at the same

37 frames, under a different or additional descriptor; Section 5 shows a stronger descriptor changes which frames connect and how, not simply how many. What is stable across both descriptors is the diagnostic itself: a single local fit's residual is not trustworthy on this dataset, full stop, and only cross-checking two structurally independent estimators against each other exposed either failure.

7. Limitations

- No ground truth exists for this sequence, so nothing here is an accuracy claim; it is a self-consistency and failure-mode analysis only.
- Translation direction was tested (Section 4.12) using a single global linear solve, the standard closed-form method; it was not cross-checked by a second, independent translation estimator the way rotation was checked against spectral synchronization, since no analogous closed-form alternative to the linear method itself exists in the same sense. The rank-deficiency check that ruled out a critical-configuration explanation was itself only run on generic synthetic motion, not on synthetic motion resembling an orbit around a fixed target, which is what this sequence actually is.
- The width-scaling correction to K (Section 2) is inferred from resolution arithmetic, not confirmed by the data source; Section 4.10 shows it does not materially change the rotation results, but it has not been independently confirmed.
- The image-statistic correlations in Section 4.3 (r around -0.5) explain roughly a quarter of the variance in tracking quality; they are real and significant but not a complete account of every failure, including the specific line-boundary case in Section 4.9.
- The vanishing-point self-calibration (Section 4.11) is an inconclusive negative result on this footage, not a demonstrated method failure in general.
- The sub-threshold cross-block rotation population in Section 4.8 (Fig. 9) is reported descriptively; its mild concentration near 160 to 180 degrees is not treated as evidence against edge (60, 82)'s 112.1 degrees, because the same population is already known, from Section 4.4, to be dominated by unreliable measurements at this inlier count.
- The multi-initialization test (Section 4.7) used 6 spanning-tree starting points; it demonstrates non-uniqueness but does not enumerate every locally optimal solution, so the true number of comparably-fitting alternative reconstructions is a lower bound, not a count.
- All thresholds (RANSAC pixel gate, ratio test, ORB feature/grid settings, the $R_H = 0.45$ planarity threshold, the 20-inlier graph threshold) are taken from established practice (ORB-SLAM, the original front end's KITTI-tuned settings) or chosen from

the sensitivity curve in Fig. 6 and Table 1, not tuned against this dataset's outcome.

8. Recommended next steps

Section 5 forecloses the most direct fix (add measurements from a stronger descriptor) and shows it does not simply reduce to picking a better feature type. The more general recommendations are what should carry forward: (1) acquire denser frame sampling (video rather than widely spaced stills) so that consecutive-frame matchability, not sparse wide-baseline luck, carries the sequence and the measurement graph is not this close to a bare path in the first place, since that is the structural root cause both descriptors ran into; (2) if a reconstruction from this exact sequence is still needed, treat neither the ORB 34-frame block nor any SIFT-based subset as pre-certified, and instead run the full cross-check (three-view pruning, then an independent second estimator, then multi-initialization or the direct NLS-versus-spectral test used in Section 5) on whatever graph the intended front end produces, since Section 5 shows this must be redone per front end rather than assumed to transfer; (3) investigate the large translation-direction residual found in Section 4.12, most directly by refining each pairwise translation direction jointly with structure (triangulated 3D points), the usual remedy when the raw two-view direction estimate is the weak part of a reconstruction, rather than by averaging the already-computed directions further; (4) confirm or replace the inferred width-scaling of K against the original capture resolution if it becomes available, though Section 4.10 shows this is unlikely to change the rotation conclusions; (5) if this sequence or a similar one is to be used for a scored assessment, obtain or construct a reference trajectory (a surveyed target, a second camera, or an independent pose source), since without one no absolute accuracy number of the kind reported for KITTI and DFKI ARIS can be produced, and no amount of internal cross-checking substitutes for one.

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